

File-System Layout

1 Basic Idea

File-System Layout

A file system is stored inside a disk partition. The partition contains special control blocks for booting, describing the file system, tracking free space, describing files, and storing ordinary directories and file data.

- A physical disk can be divided into one or more **partitions**.
- Each partition can contain an independent file system.
- The whole disk contains boot information, while each partition contains its own internal file-system layout.
- The exact layout differs between file systems, but the same basic ideas appear repeatedly.

2 Disk-Level Layout

Master Boot Record

Sector 0 of the disk is called the **MBR**, or **Master Boot Record**. It is used during system startup.

- The **MBR** contains a small boot program.
- The end of the MBR contains the **partition table**.
- The partition table records the starting and ending addresses of each partition.
- One partition is marked as **active**.
- The active partition is the partition from which the operating system is loaded.

3 Boot Sequence

1. The computer starts and the BIOS runs.
2. The BIOS reads sector 0, which contains the MBR.
3. The BIOS executes the program stored in the MBR.
4. The MBR program finds the active partition from the partition table.
5. The MBR reads the first block of the active partition.
6. This first block is the **boot block**.
7. The boot block program loads the operating system stored in that partition.

Important Detail

Every partition normally begins with a boot block, even if that partition does not currently contain a bootable operating system. This keeps partition layout uniform and allows the partition to become bootable later.

4 Typical Partition Layout

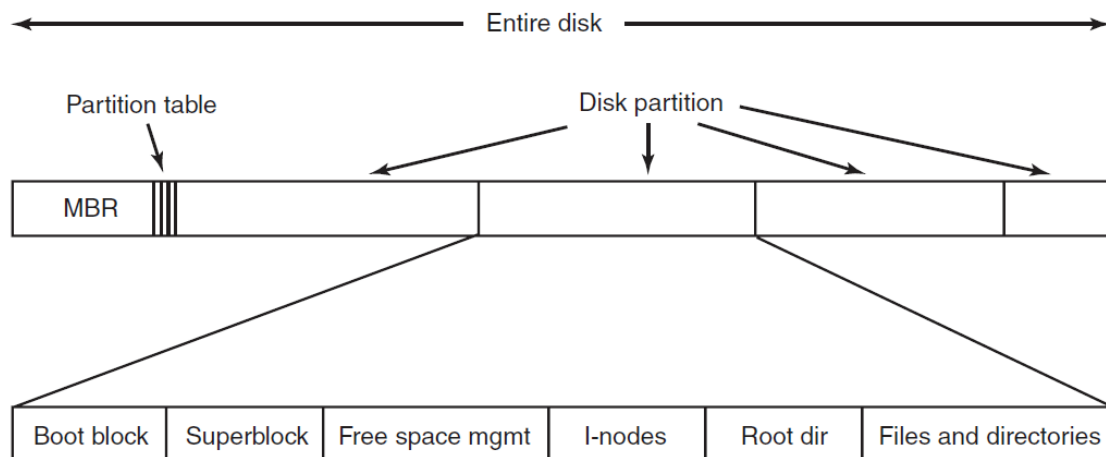


Figure 4-9. A possible file-system layout.

Figure 1: A possible file-system layout inside a disk partition.

Component	Purpose
Boot block	Contains code used to start an operating system from the partition.
Superblock	Stores key parameters that describe the file system.
Free-space management	Tracks which disk blocks are free and available for allocation.
I-nodes	Store per-file information, such as attributes and block locations.
Root directory	Stores the top-level directory of the file-system tree.
Files and directories	Store the remaining ordinary file data and subdirectories.

5 Superblock

Superblock

The **superblock** contains the main administrative information needed to understand and manage the file system.

- It is read into memory when the system boots or when the file system is first accessed.
- It usually contains a **magic number** identifying the file-system type.
- It records the number of blocks in the file system.

- It also stores other key parameters needed by the operating system.

6 Free-Space Management

- The file system must know which blocks are free before it can create or extend files.
- Free space may be represented using a **bitmap**.
- It may also be represented using a **list of pointers** to free blocks.
- Without free-space information, the system cannot safely allocate new disk blocks.

7 I-Nodes and Root Directory

- The **i-node area** is an array of data structures.
- Each i-node describes one file.
- An i-node stores information about a file rather than the file's name.
- The **root directory** is the starting point of the directory tree.
- From the root directory, the operating system can locate other directories and files.

8 Key Points

- The MBR belongs to the whole disk.
- The partition table describes where partitions begin and end.
- A boot block belongs to a partition.
- A superblock describes one file system.
- Free-space management tells the system where new data can be placed.
- I-nodes describe files.
- The root directory is the entry point into the directory hierarchy.

Final Point

A file-system layout separates boot information, file-system metadata, free-space tracking, file descriptions, and actual file data so that the operating system can boot, mount, allocate, locate, and manage files efficiently.